

RESEARCH STATEMENT

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September 2026

I am a macroeconomist studying the nexus between business cycle fluctuations and information economics. Broadly, my research seeks to understand how expectations are formed and how the resulting beliefs shape economic outcomes. This note provides an overview of my work in three parts. *Rational Inattention in Macroeconomics* studies how decision-makers allocate scarce attention across the economy, and how that allocation shapes their actions and aggregate dynamics. *Learning in Dynamic Models* turns to how agents learn about their environment from the data they observe, and what that learning implies for the transmission of shocks in general equilibrium. *Other Research* covers projects that fall into neither category.

Rational Inattention in Macroeconomics

Rational Inattention (RI) is a theory of endogenous attention introduced by [Sims \(1998, 2003\)](#). The theory centers on a fundamental trade-off that arises in any problem in which agents face uncertainty about the state of the world: acquiring more information improves decisions but raises the mental effort of doing so. RI formalizes how decision-makers deal with this cognitive limitation. Concretely, agents design signals (i.e., they choose what to pay attention to and how precisely); because attention is costly, these signals are noisy, and filtering them induces inertia in their decisions (e.g., prices, consumption, investment). In the following papers, I embed RI in different Dynamic Stochastic General Equilibrium (DSGE) models. I study how inattention helps close the gap between model predictions and the data, and how it feeds back into the propagation of macroeconomic shocks.

Inattention in Estimated Business Cycle Models

In “*Macroeconomic Shocks & Their Propagation under Rational Inattention*,” I construct, solve, and estimate a medium-scale DSGE model in which firms and households are rationally inattentive. To the best of my knowledge, my model is the first RI-DSGE to feature both pricing and capital decisions, and to be estimated with Bayesian methods.¹ One of the main results of the paper is that, in this setup, inattention can fit the data only when paired with a real friction affecting capital accumulation. The reason is easiest to see after a monetary policy shock: inattention induces slow-adjusting prices, which cause a contraction in aggregate demand; absent any other friction, the contraction produces a large drop in firms’ optimal capital stock, which draws their attention. Firms therefore track the optimal capital stock closely, so investment dynamics are peaked on impact rather than hump-shaped. General equilibrium feedback amplifies the volatility of the optimal capital stock, which only strengthens firms’ incentive to track it. General equilibrium feedback amplifies the volatility of the optimal capital stock, which only strengthens the firms’ incentive to track it. Thus, no degree of inattention alone can deliver inertia in both prices and investment.

¹[Maćkowiak and Wiederholt \(2015\)](#) feature pricing but no capital decisions; [Maćkowiak and Wiederholt \(2025\)](#) feature capital but no pricing.

I estimate the model with capital adjustment costs using quasi-Bayesian methods, by matching the impulse responses to a monetary policy shock. The model fits the evidence as well as a prototypical New Keynesian benchmark and is even favored by the marginal data density. When I feed the shock processes estimated with the New Keynesian model into the RI model, the two models nearly coincide in their responses to technology and investment shocks — untargeted, and the main drivers of the business cycle — but diverge sharply on markup shocks, long criticized as reduced-form wedges. Lastly, the RI model, in which attention is endogenous, reproduces the large, impact-peaked firm-level responses to idiosyncratic shocks documented in micro data, a pattern the New Keynesian firm cannot generate.

Inattention & Household Heterogeneity

A natural extension is to take RI to models with household heterogeneity, an active area of research. Recent theoretical work on the microeconomic transmission of monetary policy (see [Violante 2021](#) for a survey) offers a rich treatment of how consumption-savings decisions differ across the wealth distribution and how those differences decompose the aggregate response, but abstracts from heterogeneity in expectations. Endogenous attention can explain why expectations vary systematically with household characteristics, as documented in survey data ([Mitman et al. 2022](#)), and how those expectations in turn feed back into decisions. “*Monetary Policy with Inattentive Heterogeneous Households*” pursues this direction.²

I proceed as follows. First, I build a tractable heterogeneous-household model that closely matches its HANK counterpart quantile by quantile. I then follow [Auclert, Rognlie, and Straub \(2020\)](#) and introduce information frictions, which reconcile the model’s dynamics with the hump-shaped response of aggregate consumption in the data. Under homogeneous labor supply, RI and exogenous information frictions mostly agree on the decomposition of that response into direct and indirect effects, although neither easily matches the relation between expectations and wealth observed in the data. Next, I relax the assumption of homogeneous labor supply, typical of HANK models but at odds with the data ([Cantore et al. 2026](#)), and show that the two specifications then disagree. The main message of the paper is that our understanding of monetary policy transmission can change substantially once standard assumptions are relaxed, most strikingly when endogenous attention is combined with heterogeneous labor supply.

Inattention beyond Local Approximations

The models in the papers discussed above are solved in a neighborhood of the non-stochastic steady state: dynamics are linear, and agents’ losses from suboptimal actions are quadratic. In this linear-quadratic setting, Gaussian signals are optimal. The approximation, however, abstracts from important features of RI. In particular, [Matějka \(2016\)](#) shows that outside the linear-quadratic case, optimal signals are often discrete even when the state of the world is continuous. In future work, I plan to take RI to fully nonlinear macroeconomic models, where discrete signals could explain sharp movements in aggregates that fundamentals alone cannot account for.

²This paper supersedes a previously circulated version titled “*Rationally Inattentive Heterogeneous Households*.”

Learning in Dynamic Models

Learning in Heterogeneous-Agent Models

In heterogeneous-agent (HA) models with full-information rational expectations (FIRE), agents forecast prices by solving for the distribution that generates them, which is both computationally tedious and an unrealistic description of expectations formation. Moll (2025) suggests that agents should instead forecast prices directly, an approach related to Krusell and Smith (1998)’s perceived law of motion (PLM).³ In “*Non-Parametric Expectations in Heterogeneous-Agent Models*”, with Pavel Sevcik (UQAM), we pursue this idea non-parametrically: agents forecast prices and other observables with Gaussian process regression, which also quantifies how uncertain each forecast is.

We show that the resulting forecasts and aggregate dynamics differ most from those under standard PLMs in two cases: in finite samples, where posterior variance remains wide in rarely visited regions of the state space, and when the aggregate state cannot be recovered from the observables. The consequences are ambiguous: posterior variance above its irreducible level raises precautionary savings, whereas confounding shocks, for instance transitory and persistent ones, can push the other way. We argue that our method better represents how agents form expectations and delivers economically interesting behavior.

Sparse Learning

Models of sparse learning (Gabaix 2014, 2016), in which agents simplify their world and decisions by shrinking the weights they place on different state variables, have seen few macroeconomic applications. Yet the framework is naturally close to more complex and more established models, such as RI, in which sparsity emerges from the optimal allocation of attention, while being far cheaper to solve. In a paper tentatively titled “*Learning under Constraint: Implications for Macroeconomics*”, with Sergey Slobodyan (CERGE-EI), we show that under certain assumptions sparse learning is equivalent, or very close, to RI, so that RI can be read as a form of sparse learning. We then exploit this correspondence to estimate DSGE models in which learning is costly. The framework also offers more flexibility: dynamics can reflect agents learning about the economy’s laws of motion over time, not only about its current state.

Other Research

News and Noise in Expectations

Expectations about future productivity are a natural source of business cycle fluctuations, but they need not reflect fundamentals alone: agents receive genuine information about future productivity (news) as well as signals unrelated to it (noise). The literature disagrees on how much noise matters. In “*Noisy Expectations and Business Cycles: Identification with Local Projections*”, with Alain Guay (UQAM) and Patrick Fève (TSE), we trace that disagreement to two sources. We first show that existing procedures exploit different information sets and therefore need not recover the same object. We characterize when a structural impulse

³The idea is close to how RI models are solved.

response can be recovered from a finite window of future observations, and develop a simple local-projection framework that recovers, on the same data, the responses to either the Wold innovation of Total Factor Productivity (TFP) growth or the structural news and noise shocks. We then argue that how noisy expectations are and how much noise shocks matter for fluctuations are distinct quantities, and propose long-run statistics that measure the news content of expectation signals. Applying these tools to U.S. data, we find that roughly one third of the new information in consumer-confidence signals is noise, yet noise shocks have only small and transitory effects on real activity, while news shocks generate persistent expansions. We conclude that expectations can be substantially noisy without noise being an important source of business cycle fluctuations.

Attention and Inflation Dynamics

Attention to inflation is not constant: it rises when inflation is high and is shaped by media coverage, which frames the information agents act on. In “*Inflation, Attention and Expectations*”, with Massimiliano Marcellino (Bocconi) and Dalibor Stevanovic (UQAM), we measure attention with Google Trends searches for “inflation” and the share of *Wall Street Journal* headlines about inflation. We show that both measures explain inflation expectations, suggesting that attention affects inflation through expectations. To rationalize this channel, we build a dispersed-information price-setting model in which firms use a public signal only when attention is high, which anchors their higher-order beliefs and mutes the real effects of monetary policy. Estimating a VAR separately on low- and high-attention periods, we show that the real effects of monetary policy are indeed smaller when attention is high.

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